3.8 Digital Processing of Continuous-Time Signals

- *Sampling of Continuous-Time Signals*
- *Recovery of the Analog Signal*
- *Implications of the Sampling Process*

3.9 Sampling of Band-pass Signals

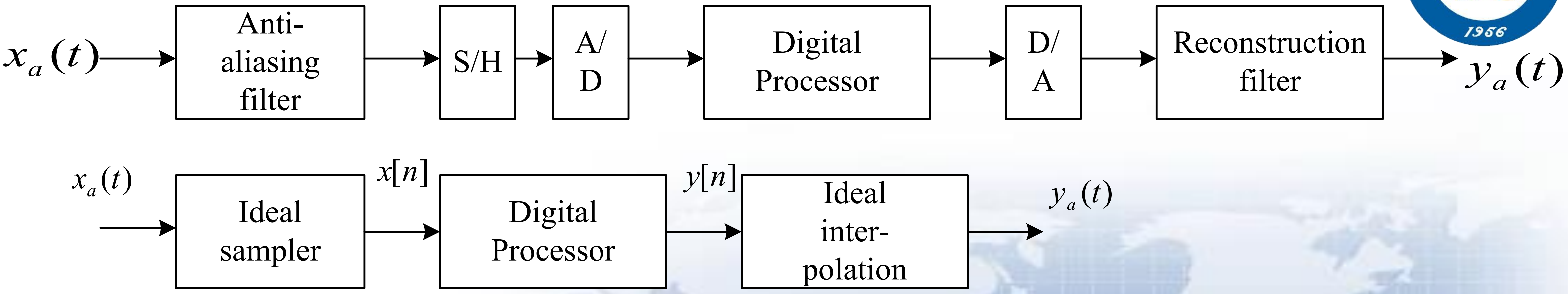


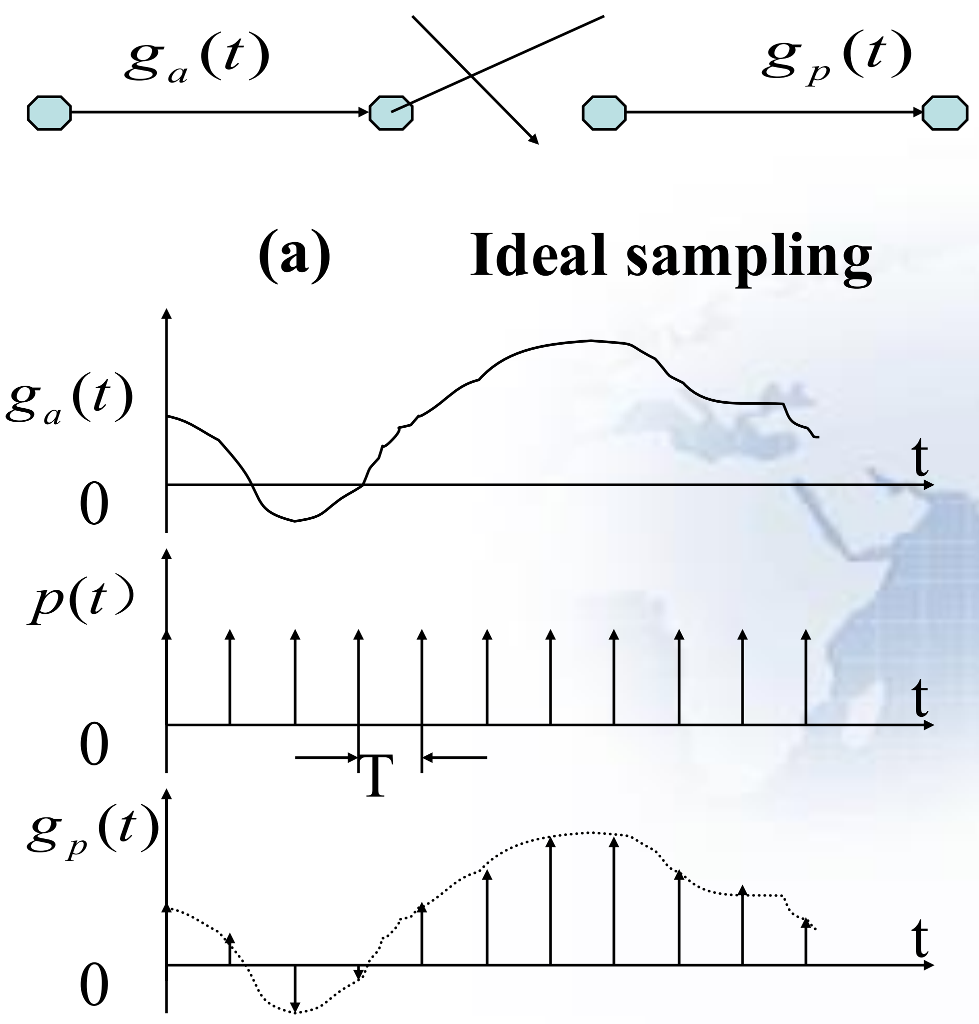
Fig 4.1,4.2 Digital Processing of A Continuous Signal

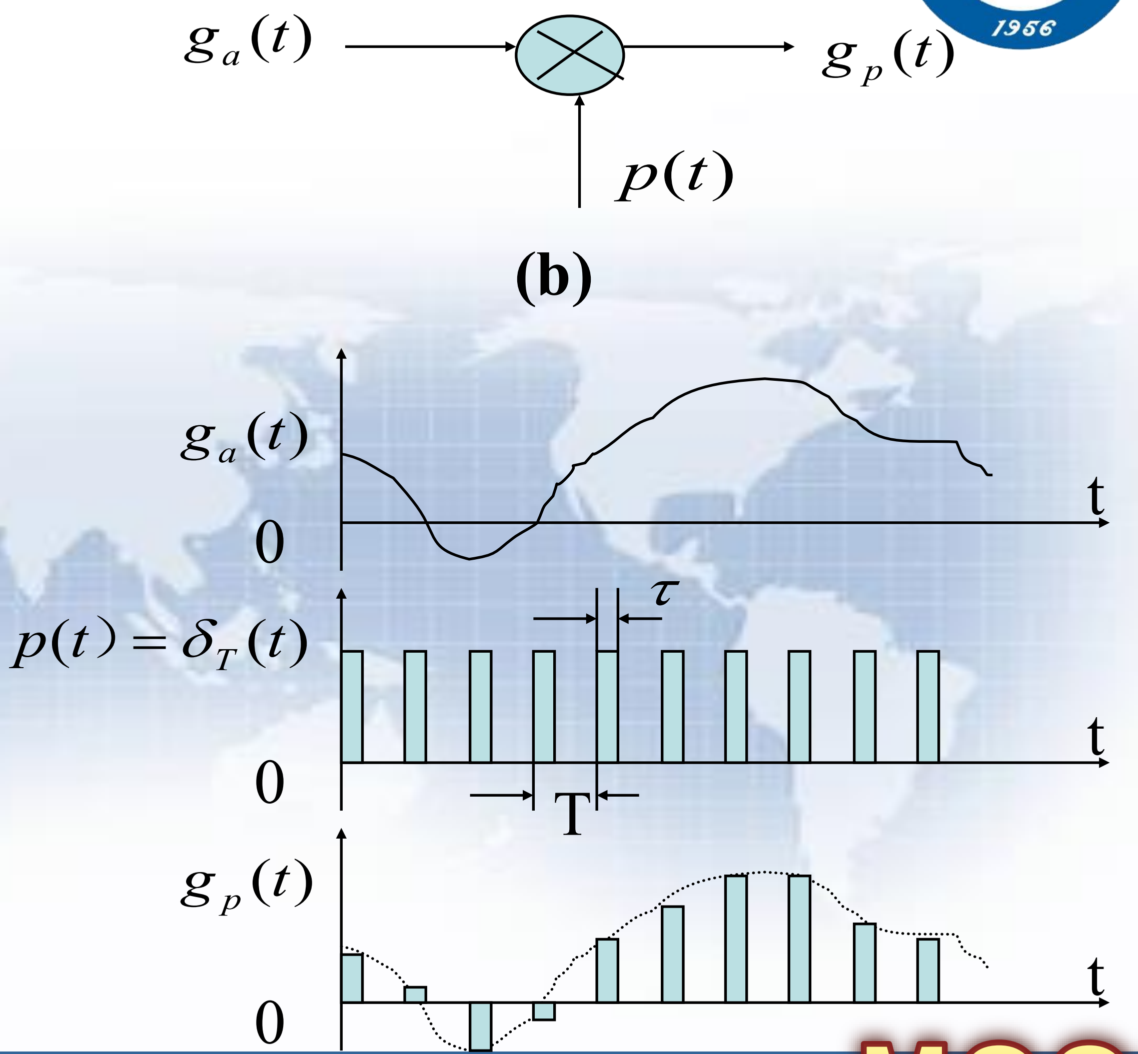
Some concepts:

1. **ADC: Analog-to-Digital Converter**
2. **DAC: Digital-to-Analog Converter**
3. **S/H : Sample-and-Hold (circuit).**
4. **Anti-aliasing filter.**
5. **Reconstruction or interpolation filter.**

MOOC

3.8.1 Effect of Sampling in the Frequency-Domain





- **When τ« T, *p*(t) ~δ_T(t);**

so we firstly discuss ideal sampling:

$$\mathrm{Q}\ \delta_T(\mathrm{t}) = \sum_{n=-\infty}^{\infty} \delta(t - nT)$$

The output of ideal sampling is:

(1)In time domain:

$$g_p(t) = g_a(t) g \delta_T(t) \quad = \sum_{n=-\infty}^{\infty} g_a(nT)\delta(t - nT)$$

(2)In frequency domain:

$$G_p(j\Omega) = \frac{1}{2\pi} G_a(j\Omega) * \Delta_T(j\Omega) = \frac{1}{T}\sum_{k=-\infty}^{\infty} G_a\left(j(\Omega - k\Omega_T)\right)$$

$$= \frac{1}{T}\sum_{k=-\infty}^{\infty} G_a\left(j(\Omega - k\frac{2\pi}{T})\right) \qquad \Omega_T = \frac{2\pi}{T}$$

MOOC

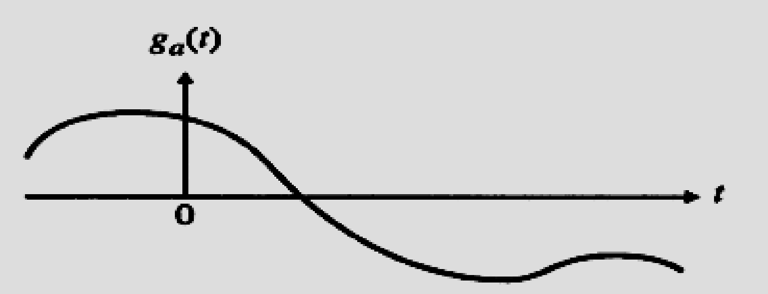
- **There are two different forms of $G_p(j\Omega)$:**

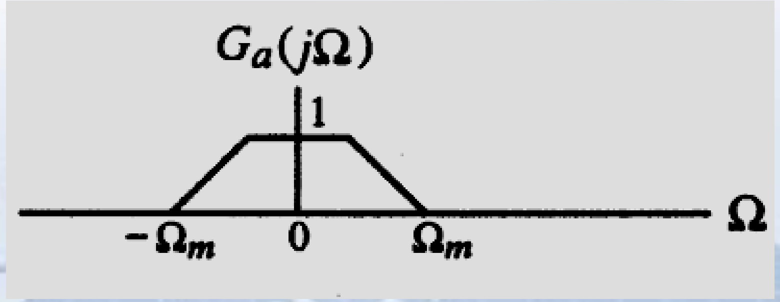
$$(1)G_p(j\Omega)=\sum_{n=-\infty}^{\infty} g_a(nT)e^{-j\Omega nT}$$

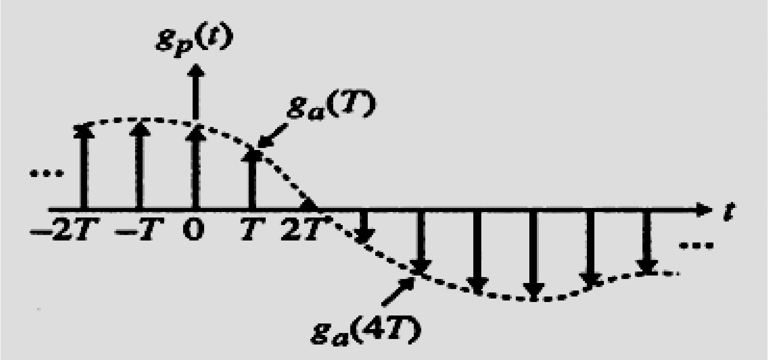
$$(2)G_p(j\Omega)=\frac{1}{T}\sum_{k=-\infty}^{\infty} G_a\left(j(\Omega-k\Omega_T)\right)$$

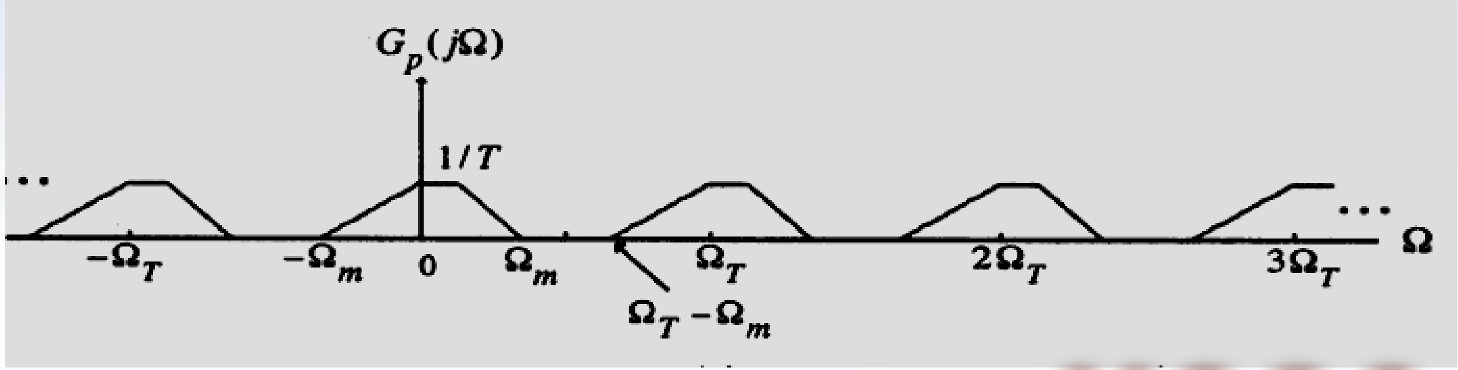
$G_p(j\Omega)$ is a periodic function of Ω consisting of a sum of shifted and scaled replicas of $G_a(j\Omega)$, shifted by integer multiples of Ω_T and scaled by 1/T.

The Discretization In time Domain ⟷ The periodization In Frequency Domain



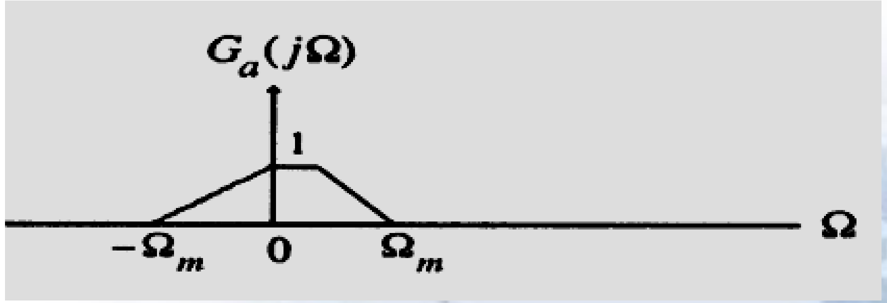




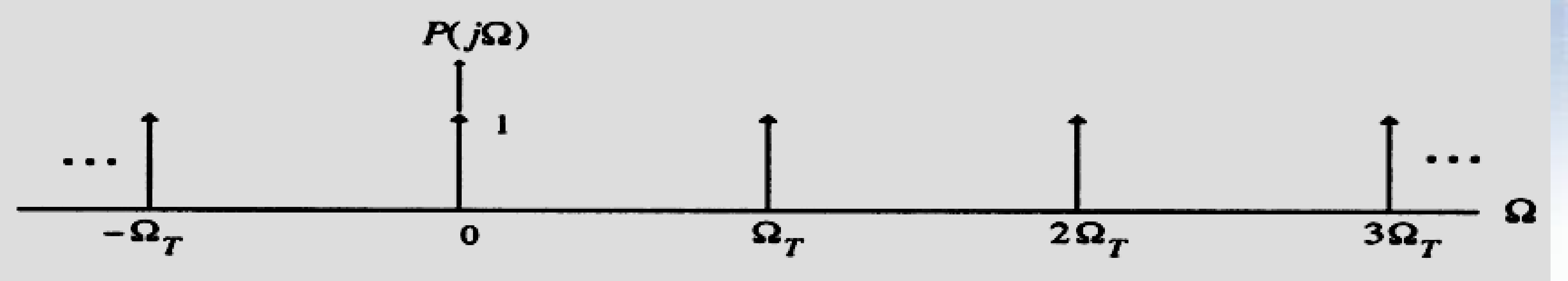


□ The frequency procedure of sampling

- Assume $g_a(t)$ is a band-limited signal with a CTFT $G_a(j\Omega)$ as shown below:

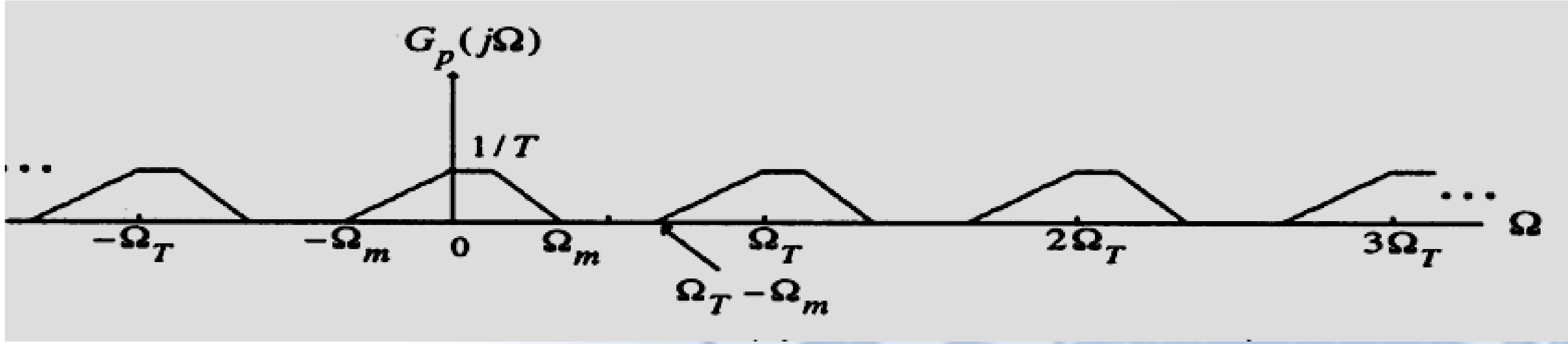


- The spectrum $P(j\Omega)$ of $p(t)$ having a sampling period $T=2\pi/\Omega_T$ is indicated below:



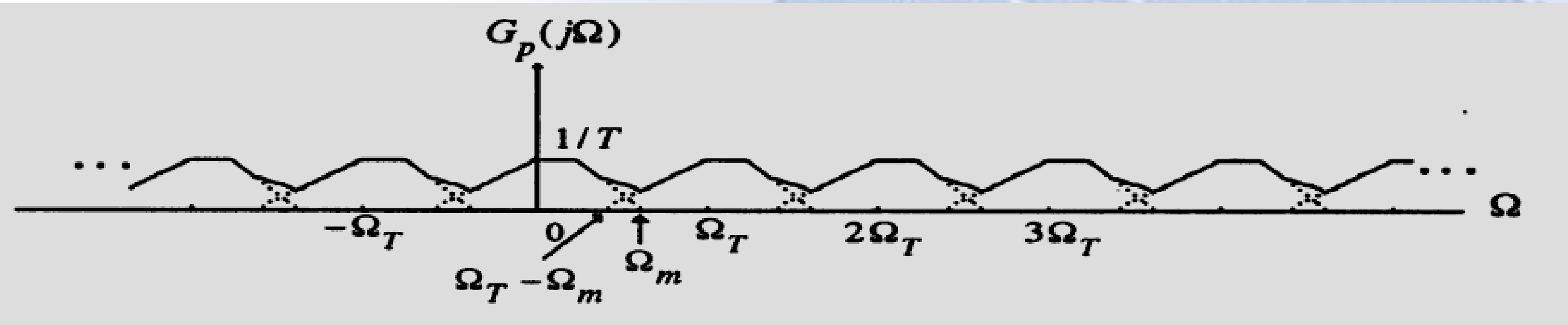
MOOC

- **Two possible spectra of $G_p(j\Omega)$ are shown below:**



$\Omega_T > 2\ \Omega_m$

no overlap



$\Omega_T < 2\ \Omega_m$

overlap/ aliasing

➢ Sampling Theorem

- Let $g_a(t)$ be a band-limited signal with CTFT $G_a(j\Omega)=0$ for $|\Omega| > \Omega_m$,

  Then $g_a(t)$ is uniquely determined by its samples $g_a(nT)$, $-\infty \leq n \leq \infty$ if

$$\Omega_T \geq 2\,\Omega_m, \quad \text{where } \Omega_T = 2\pi/T$$

- The condition $\Omega_T \geq 2\,\Omega_m$ is often referred to as the **Nyquist condition.**
- The frequency $\Omega_T/2$ is usually referred to as the **folding frequency.**

MOOC

- The highest frequency Ω_m contained in $g_a(t)$ is usually called the Nyquist frequency since it determines the minimum sampling frequency $\Omega_T = 2\Omega_m$ that must be used to fully recover $g_a(t)$ from its sampled version.
- The frequency $2\Omega_m$ is called the Nyquist rate.

MOOC

- **Oversampling** - The sampling frequency is higher than the Nyquist rate;
- **Undersampling** - The sampling frequency is lower than the Nyquist rate;
- **Critical sampling** - The sampling frequency is equal to the Nyquist rate;
- **Note:** A pure sinusoid may not be recoverable from its critically sampled version.

Nyquist frequency Examples:

- In **digital telephony**, a 3.4 kHz signal bandwidth is acceptable for telephone conversation;
- Here, a sampling rate of 8 kHz, which is greater than twice the signal bandwidth, is used.
- In **high-quality music signal processing**, a bandwidth of 20 kHz has been determined to preserve the fidelity;
- Hence, in compact disc (CD) music systems, a sampling rate of 44.1 kHz, which is slightly higher than twice the signal bandwidth, is used.

The Relationship Between $G(e^{j\omega})$ *and* $G_p(j\Omega)$

- We now derive the relation between the DTFT of g[n] and the CTFT of $g_p(t)$
- To compare:

$$G(e^{j\omega}) = \sum_{n=-\infty}^{\infty} g[n]e^{-j\omega n}$$

with

$$G_p(j\Omega) = \sum_{n=-\infty}^{\infty} g_a(nT)e^{-j\Omega nT}$$

And make use of $g[n]=g_a(nT)$, $-\infty<n<\infty$

MOOC

The Relationship Between $G(e^{j\omega})$ *and* $G_p(j\Omega)$

- **Observation: We have**

$$G(e^{j\omega})=G_p(j\Omega)|_{\Omega=\omega/T}$$

Or, equivalently,

$$G_p(j\Omega)= G(e^{j\omega})|_{\omega=\Omega T}$$

From the above observation and

$$G_p(j\Omega)=\frac{1}{T}\sum_{k=-\infty}^{\infty}G_a(j(\Omega-k\Omega_T))$$

MOOC

The Relationship Between G($e^{j\omega}$) *and* $G_p(j\Omega)$

- **We arrive at the desired result:**

$$G(e^{j\omega}) = \frac{1}{T}\sum_{k=-\infty}^{\infty} G_a(j\Omega - jk\Omega_T)\,|_{\Omega=\omega/T}$$

$$= \frac{1}{T}\sum_{k=-\infty}^{\infty} G_a(j\frac{\omega}{T} - jk\Omega_T)$$

$$= \frac{1}{T}\sum_{k=-\infty}^{\infty} G_a(j\frac{\omega}{T} - j\frac{2\pi k}{T})$$

MOOC

The Relationship Between $G(e^{j\omega})$ *and* $G_p(j\Omega)$

- **The relation derived on the previous slide can be alternately expressed as**

$$G(e^{j\Omega T}) = \frac{1}{T}\sum_{k=-\infty}^{\infty} G_a(j\Omega - jk\Omega_T)$$

From $$G(e^{j\omega}) = G_p(j\Omega)\big|_{\Omega=\omega/T}$$

Or from $$G_p(j\Omega) = G(e^{j\omega})\big|_{\omega=\Omega T}$$

It follows that $G(e^{j\omega})$ is obtained from $G_p(j\Omega)$ by applying the mapping $\Omega=\omega/T$

MOOC

- Now, the CTFT $G_p(j\,\Omega)$ is a periodic function of Ω with a period $\Omega_T = 2\pi/T$

- Because of the mapping, the DTFT $G(e^{j\omega})$ is a periodic function of ω with a period 2π

MOOC

Supplement: The practical sampling

- **In practical condition, p(t) is a periodic rectangle impulse with τ-width .**

$$p(t)=\sum_{k=-\infty}^{\infty}C_k e^{jk\Omega_s t}$$

$$C_k=\frac{1}{T}\int_{-T/2}^{T/2}p(t)e^{-jk\Omega_s t}dt=\frac{1}{T}\int_0^{\tau}e^{-jk\Omega_s t}dt=\frac{\tau}{T}\frac{\sin(\frac{k\Omega_s\tau}{2})}{\frac{k\Omega_s\tau}{2}}e^{-j\frac{k\Omega_s\tau}{2}}$$

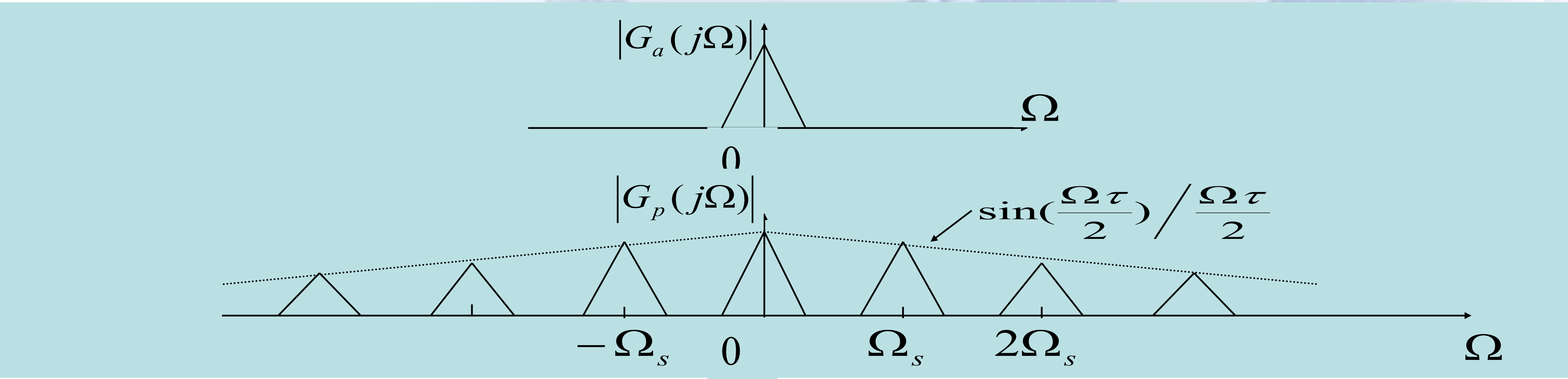
If the τ and T are fixed, then when k varies, the magnitude of C_k varies with:

$$\left|\frac{\sin(\frac{k\Omega_s\tau}{2})}{\frac{k\Omega_s\tau}{2}}\right|$$

Supplement: The practical sampling

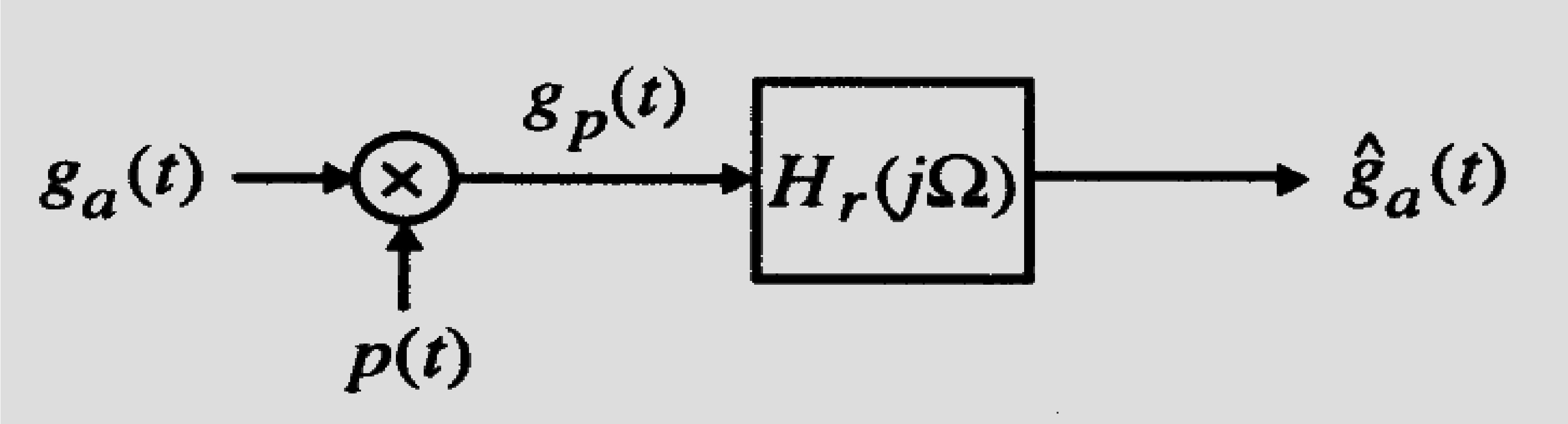
- **So we can see, in practical sampling, use C_k to substitute A_k, but C_k is varying.**

$$\therefore G_p(j\Omega) = \sum_{k=-\infty}^{\infty} C_k G_a(j\Omega - jk\Omega_s)$$



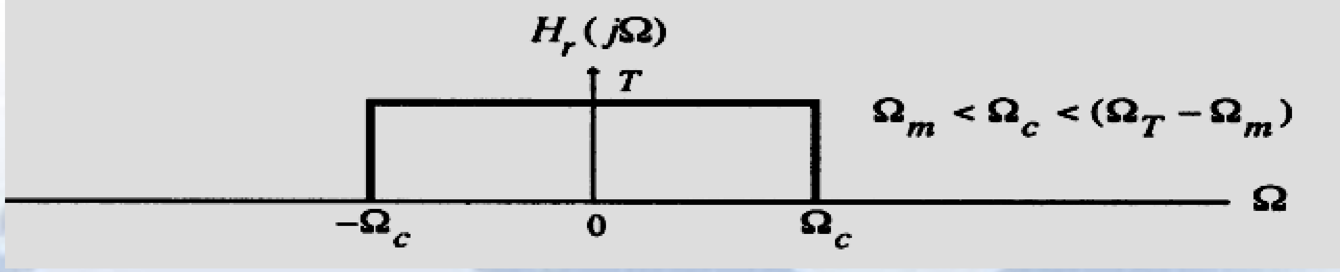
3.8.2 Recovery of the Analog Signal

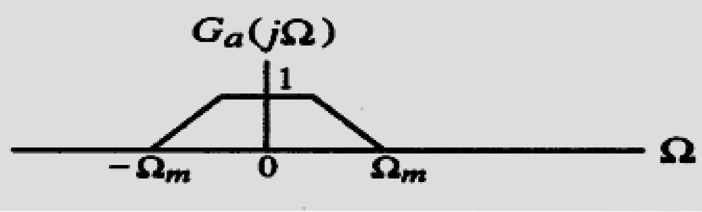
By filtering in frequency domain or interpolation in time domain, the continuous signal can be recovered from *The Sampled Discrete time signal.*

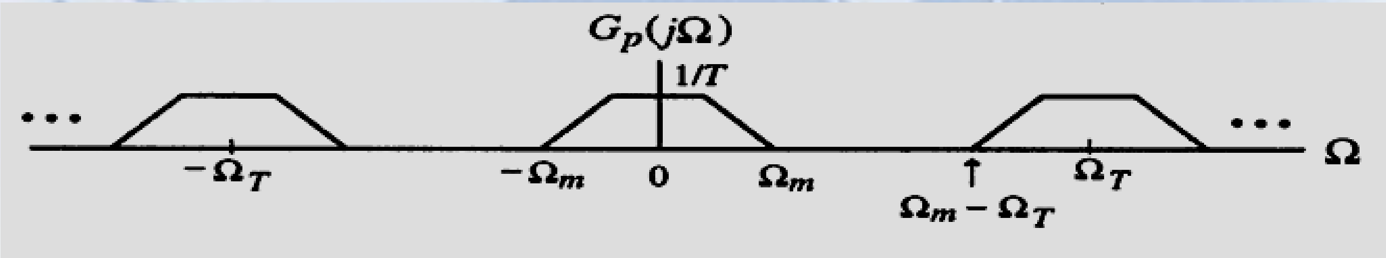


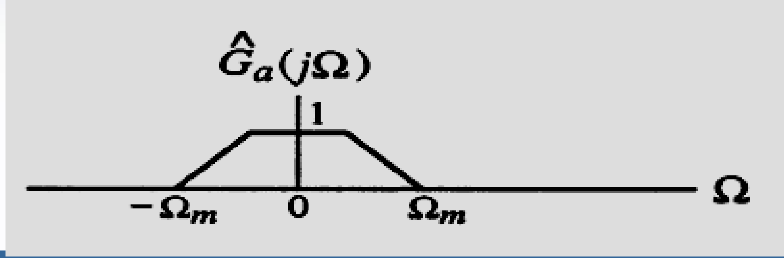
- **Here, the filter is called *reconstruction filter* or *interpolation filter* or *smoothing filter*.**

- **The spectra of the filter and pertinent signals are shown below:**









- **If $\Omega_T > 2\,\Omega_m$, $g_a(t)$ can be recovered exactly from $g_p(t)$ by passing it through an ideal lowpass filter $H_r(j\Omega)$ with a gain T and a cutoff frequency Ω_c greater than Ω_m and less than $\Omega_T - \Omega_m$;**

- **On the other hand, if $\Omega_T < 2\,\Omega_m$, due to the overlap of the shifted replicas of $G_a(j\Omega)$, the spectrum $G_p(j\Omega)$ cannot be separated by filtering to recover $G_a(j\Omega)$. The distortion caused by a part of the replicas outside the baseband folded back or aliased into the baseband.**

- **We now derive the expression for the output $\hat{g}_a(t)$ of the ideal lowpass reconstruction filter H$_r$(jΩ) as a function of the samples *g*[*n*].**

- **The impulse response h$_r$(t) of the ideal lowpass reconstruction filter is obtained by taking the inverse DTFT of H$_r$(jΩ)**

$$H_r(j\Omega)=\begin{cases} T, & |\Omega|\le\Omega_c \\ 0, & |\Omega|>\Omega_c \end{cases}$$

- **Thus, the impulse response is given by**

$$h_r(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} H_r(j\Omega)e^{j\Omega t}d\Omega = \frac{T}{2\pi}\int_{-\Omega_c}^{\Omega_c} e^{j\Omega t}d\Omega$$

$$= \frac{\sin(\Omega_c t)}{\Omega_T t/2}, \qquad -\infty \le t \le \infty$$

- **The input to the lowpass filter is the impulse train $g_p(t)$:**

$$g_p(t) = \sum_{n=-\infty}^{\infty} g[n]\delta(t-nT)$$

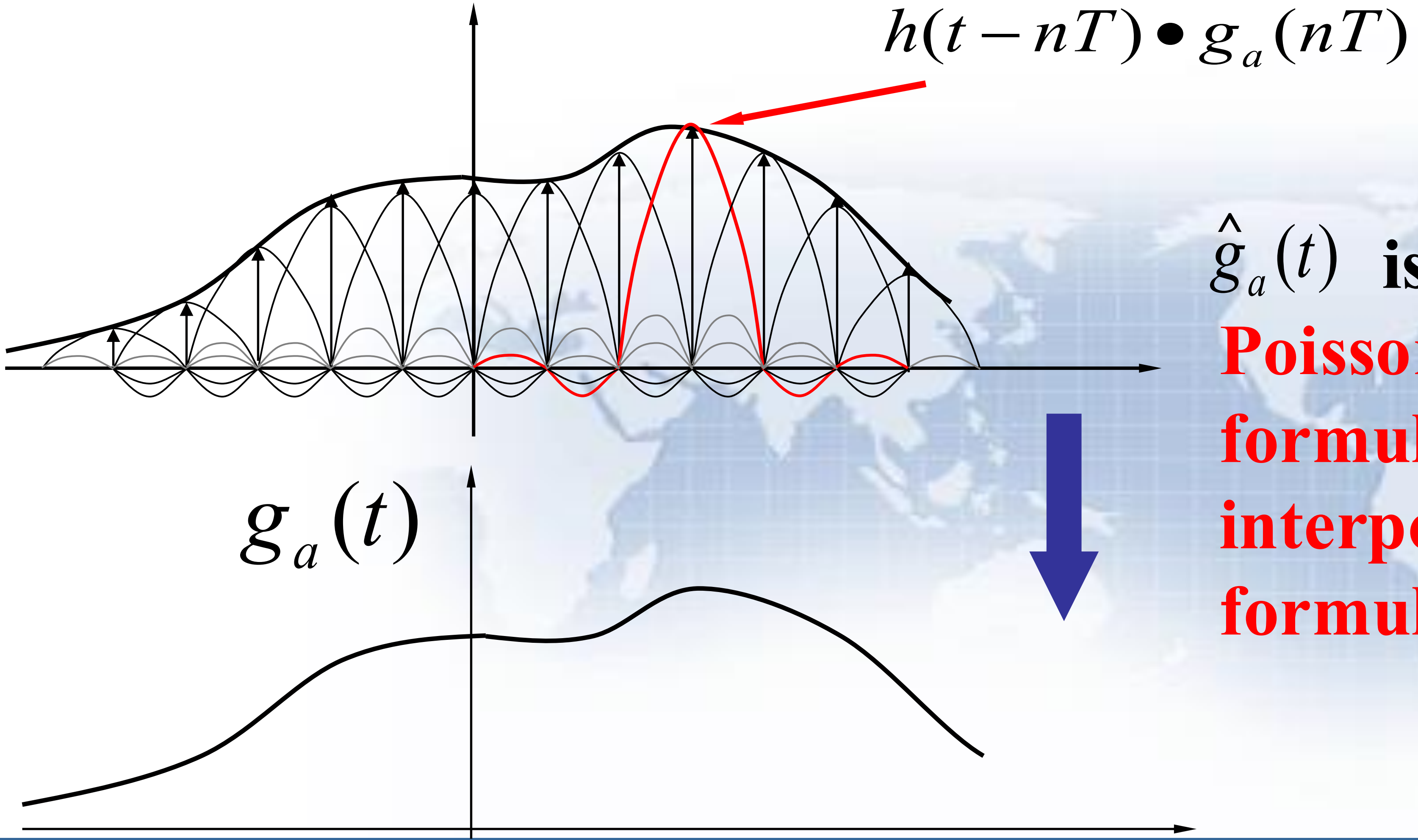
- Therefore, the output $\hat{g}_a(t)$ of the ideal lowpass filter is given by:

$$\hat{g}_a(t) = h_r(t) \circledast g_p(t) = \sum_{n=-\infty}^{\infty} g[n] h_r(t-nT)$$

Applying $h_r(t)=\sin(\Omega_c t)/(\Omega_T t/2)$, and assuming:

$\Omega c = \Omega_T/2 = \pi/T$, we get:

$$\hat{g}_a(t) = \sum_{n=-\infty}^{\infty} g[n] \frac{\sin[\pi(t-nT)/T]}{\pi(t-nT)/T}$$



$\hat{g}_a(t)$ **is called**

Poisson sum formula or interpolation formula.

3.8.3 Implications of the Sampling Process

- **<u>Example:</u> Consider the three continuous time sinusoidal signals:**

$$g_1(t) = \cos(6\pi t)$$

$$g_2(t) = \cos(14\pi t)$$

$$g_3(t) = \cos(26\pi t)$$

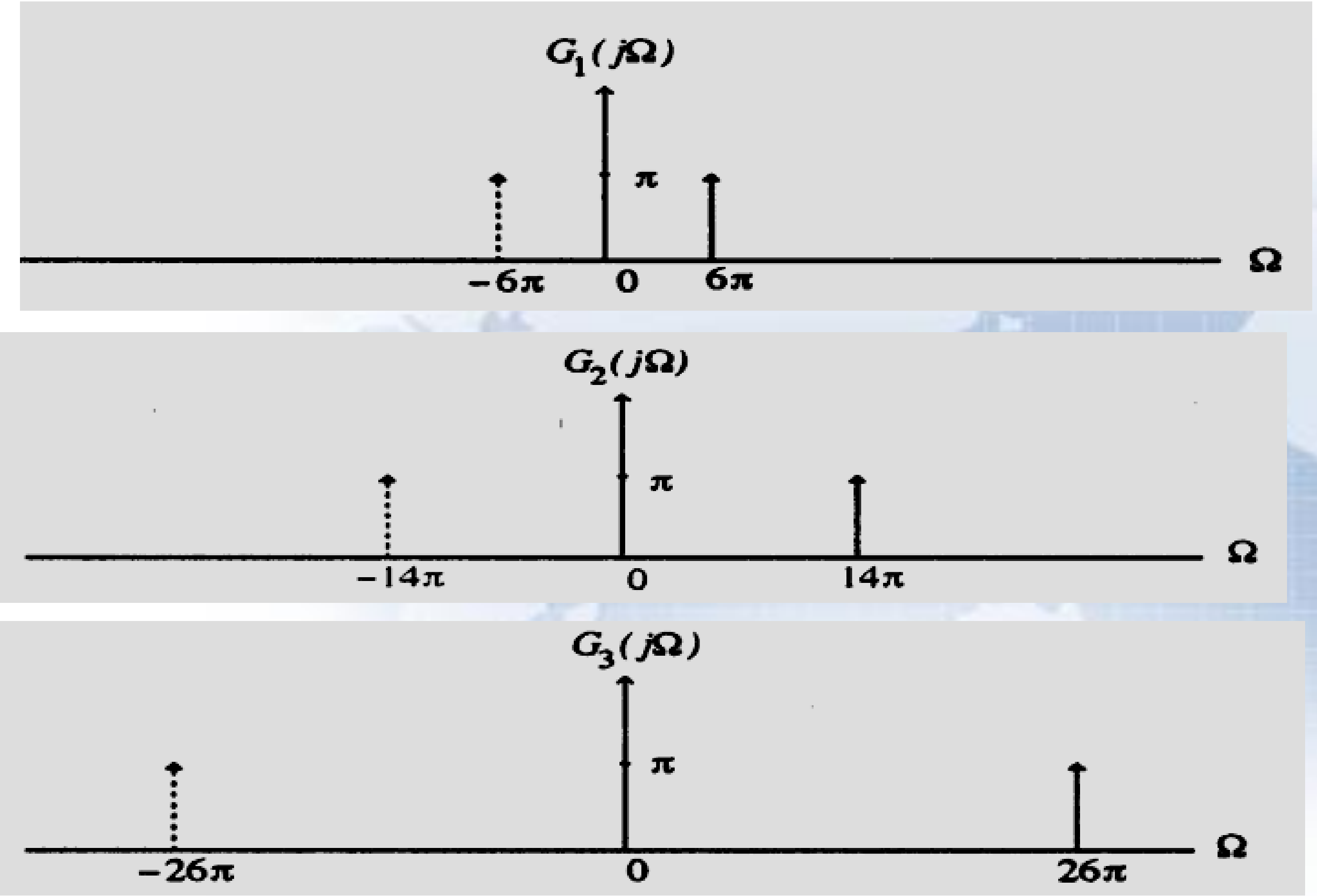
- **Their corresponding CTFTs are:**

$$G_1(j\Omega) = \pi[\delta(\Omega - 6\pi) + \delta(\Omega + 6\pi)]$$

$$G_2(j\Omega) = \pi[\delta(\Omega - 14\pi) + \delta(\Omega + 14\pi)]$$

$$G_3(j\Omega) = \pi[\delta(\Omega - 26\pi) + \delta(\Omega + 26\pi)]$$

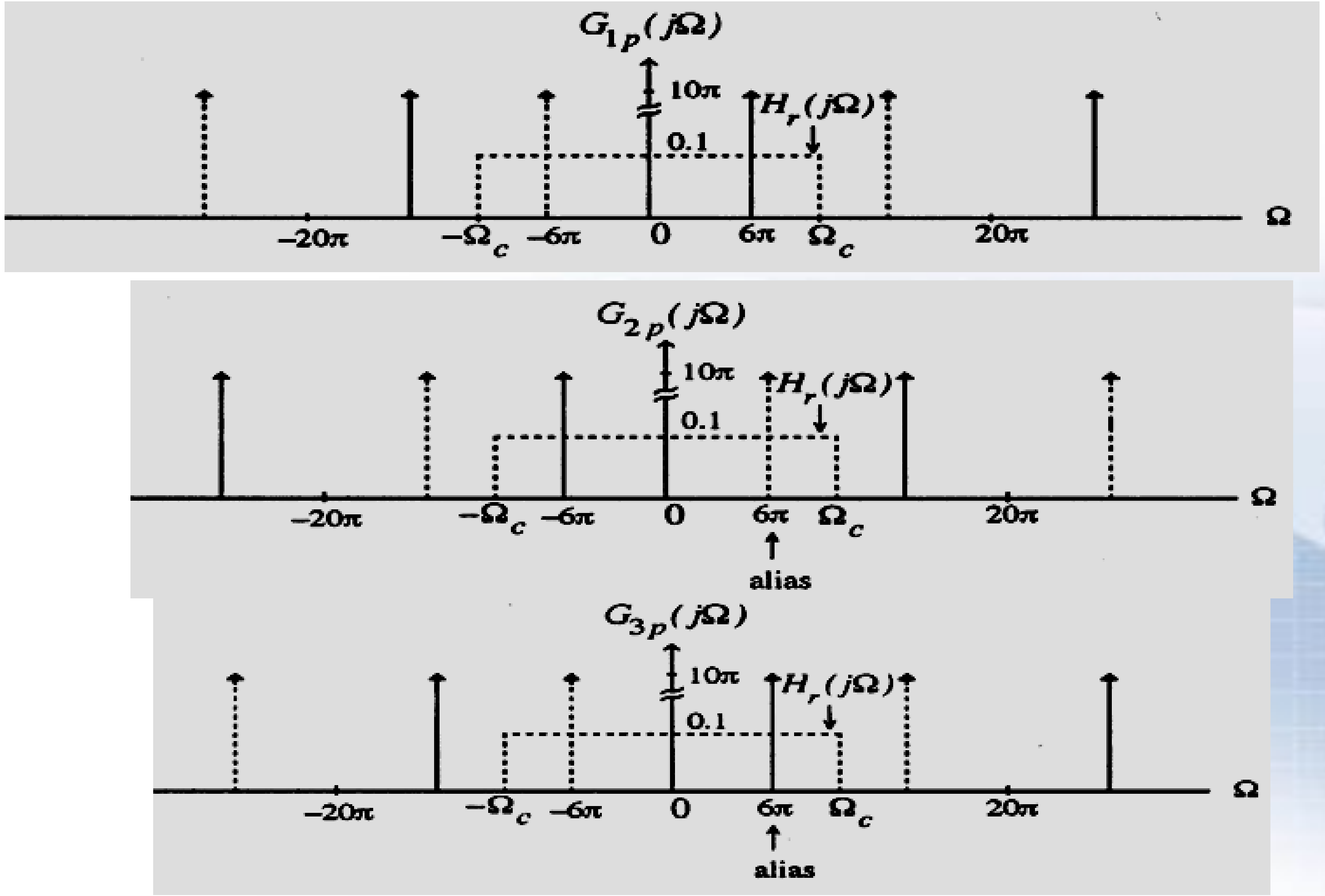
- **These three transforms are plotted below**



- These continuous-time signals sampled at a rate of T = 0.1 sec, i.e., with a sampling frequency $\Omega_T = 20\pi$ rad/sec
- The sampling process generates the continuous-time impulse trains, $g_{1p}(t)$, $g_{2p}(t)$, and $g_{3p}(t)$
- Their CTFTs are given by:

$$G_{\ell p}(j\Omega) = 10\sum_{k=-\infty}^{\infty} G_\ell(j(\Omega - k\Omega_T)), \quad 1 \le \ell \le 3$$

- **Plots of the 3 CTFTs are shown below:**



$$\{\Omega_k\} = \{\pm\Omega_0 \pm k\Omega_T\} = \{\pm 6\pi \pm 20\pi \times k\}$$

- **These figures also indicate by dotted lines the frequency response of an ideal lowpass filter with a cutoff at $\Omega_c=\Omega_T/2=10\pi$ and a gain T=0.1.**
- **In fact, the 3 discrete-time sinusoidal signals are:**

$$g_1[n] = g_1(nT) = \cos(6\pi n/10) = \cos(0.6\pi n)$$

$$g_2[n] = g_2(nT) = \cos(14\pi n/10) = \cos(1.4\pi n)$$

$$g_3[n] = g_3(nT) = \cos(26\pi n/10) = \cos(2.6\pi n)$$

And $$g_1[n] = g_2[n] = g_3[n]$$

- **If reconstruction filter**

$$H_r(j\Omega)=\begin{cases}0.1, & (6-\Delta)\pi\le|\Omega|\le(6+\Delta)\pi\\ 0, & otherwise,\end{cases}$$

$$\hat{g}_a(t)=\cos(6\pi t)$$

- **If reconstruction filter**

$$H_r(j\Omega)=\begin{cases}0.1, & (34-\Delta)\pi\le|\Omega|\le(34+\Delta)\pi\\ 0, & otherwise,\end{cases}$$

$$\hat{g}_a(t)=\cos(34\pi t)$$

3.9 Sampling of Band-pass Signals

- The conditions developed earlier for the unique representation of a continuous-time signal by the discrete-time signal obtained by uniform sampling assumed that the continuous-time signal is bandlimited in the frequency range from DC to Ω_m --- lowpass signal.
- There are continuous-time signals which are bandlimited to a higher frequency range $\Omega_L \leq |\Omega| \leq \Omega_H$ with $\Omega_L > 0$ --- bandpass signals.

- To prevent aliasing a bandpass signal can of course be sampled at a rate greater than twice the highest frequency, i.e. by ensuring $\Omega_T \geq 2\,\Omega_H$
- However, due to the bandpass spectrum of the continuous-time signal, the spectrum of the discrete-time signal obtained by sampling will have spectral gaps with no signal components present in these gaps.
- Moreover, if Ω_H is very large, the sampling rate also has to be very large which may not be practical in some situations.

- Let $\Delta\Omega = \Omega_H - \Omega_L$ define the bandwidth of the bandpass signal.
- Assume: $\Omega_H = M(\Delta\Omega)$
- We choose the sampling frequency Ω_T to satisfy the condition:

$$\Omega_T = 2(\Delta\Omega) = 2\Omega_H/M$$

- Ω_T is smaller than $2\Omega_H$, the Nyquist rate

undersampling

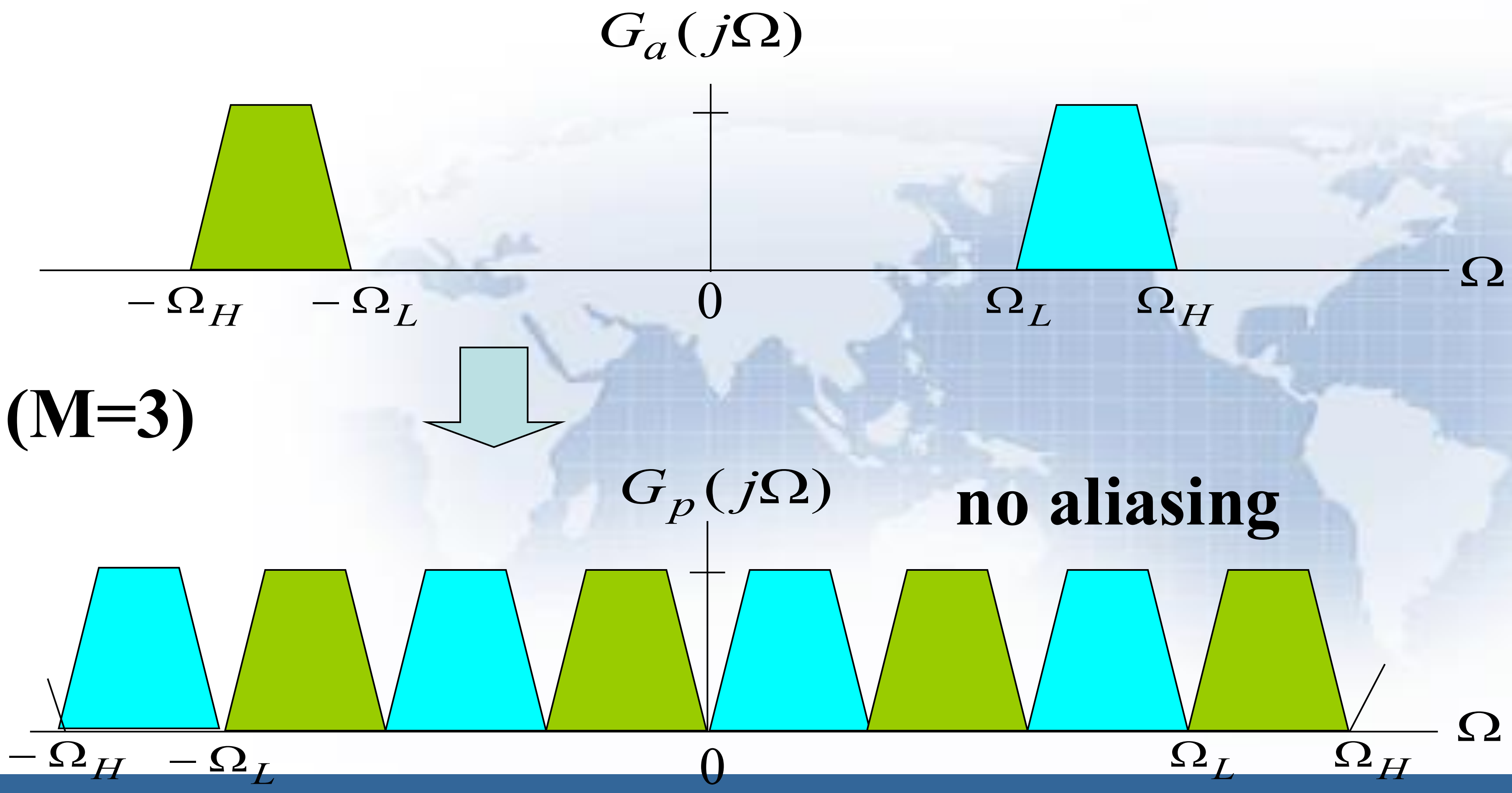
$$G_p(j\Omega) = \frac{1}{T}\sum_{k=-\infty}^{\infty} G_a(j(\Omega - k\Omega_T))$$

$$= \frac{1}{T}\sum_{k=-\infty}^{\infty} G_a\left(j\Omega - j2k(\Delta\Omega)\right)$$

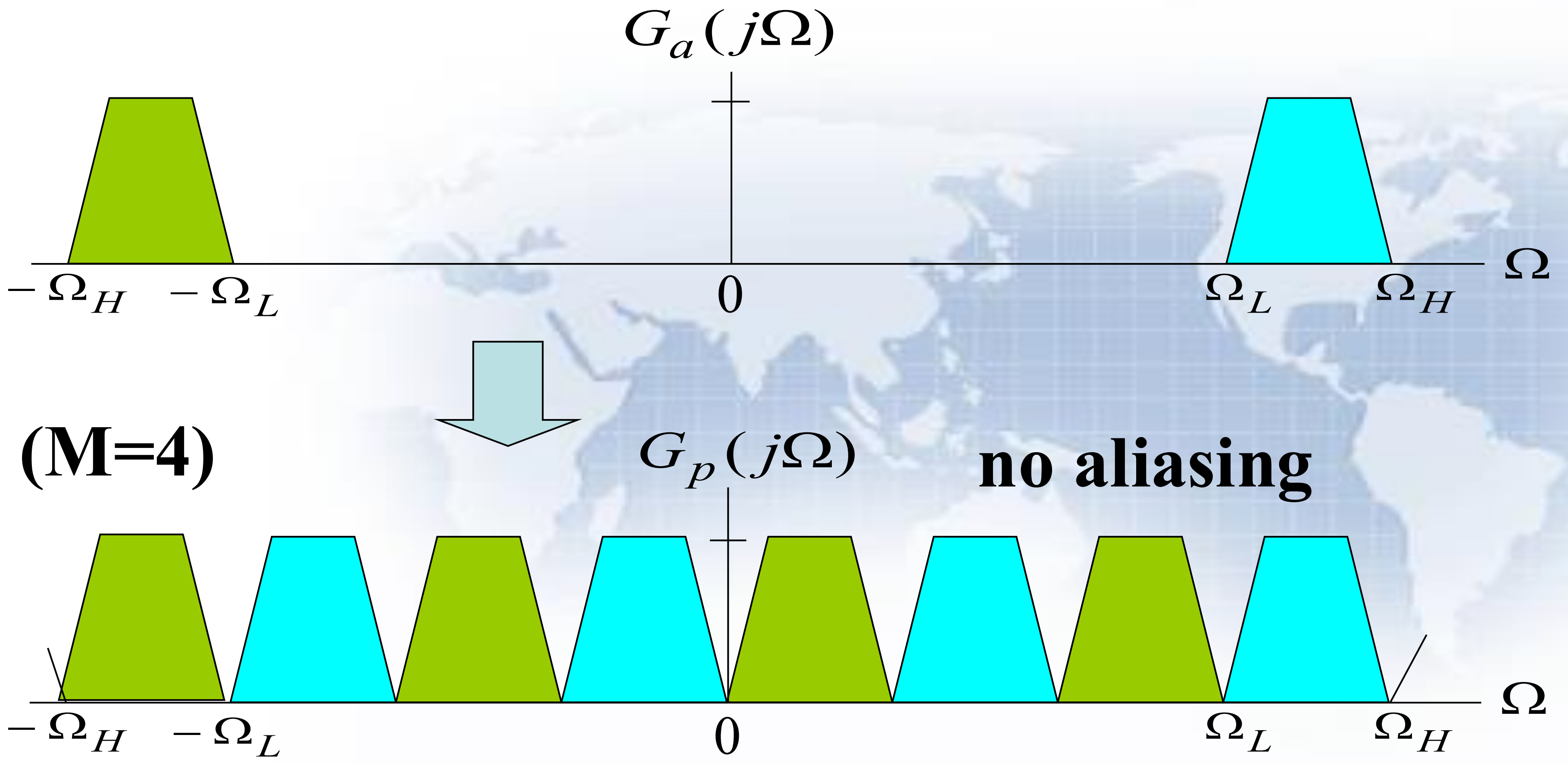
- As before, $G_p(j\Omega)$ consists of a sum of $G_a(j\Omega)$ and replicas of $G_p(j\Omega)$ shifted by integer multiples of twice the bandwidth $\Delta\omega$ and scaled by 1/T;
- The amount of shift for each value of k ensures that there will be **no overlap** between all shifted replicas.

no aliasing

- **Figures below illustrate the idea:**



- **Figures below illustrate the idea:**



Discuss:

(1)If $g_a(t)$ is bandlimited with its frequency band in (f_L, f_H) .

(2)If using sampling rate:

$$f_s = \frac{2(f_L + f_H)}{2n+1} = \frac{4f_0}{2n+1}$$

Where, n is a maximum integer which meets $fs \geq 2(f_H - f_L) = 2B$ (n = 0,1,2……)

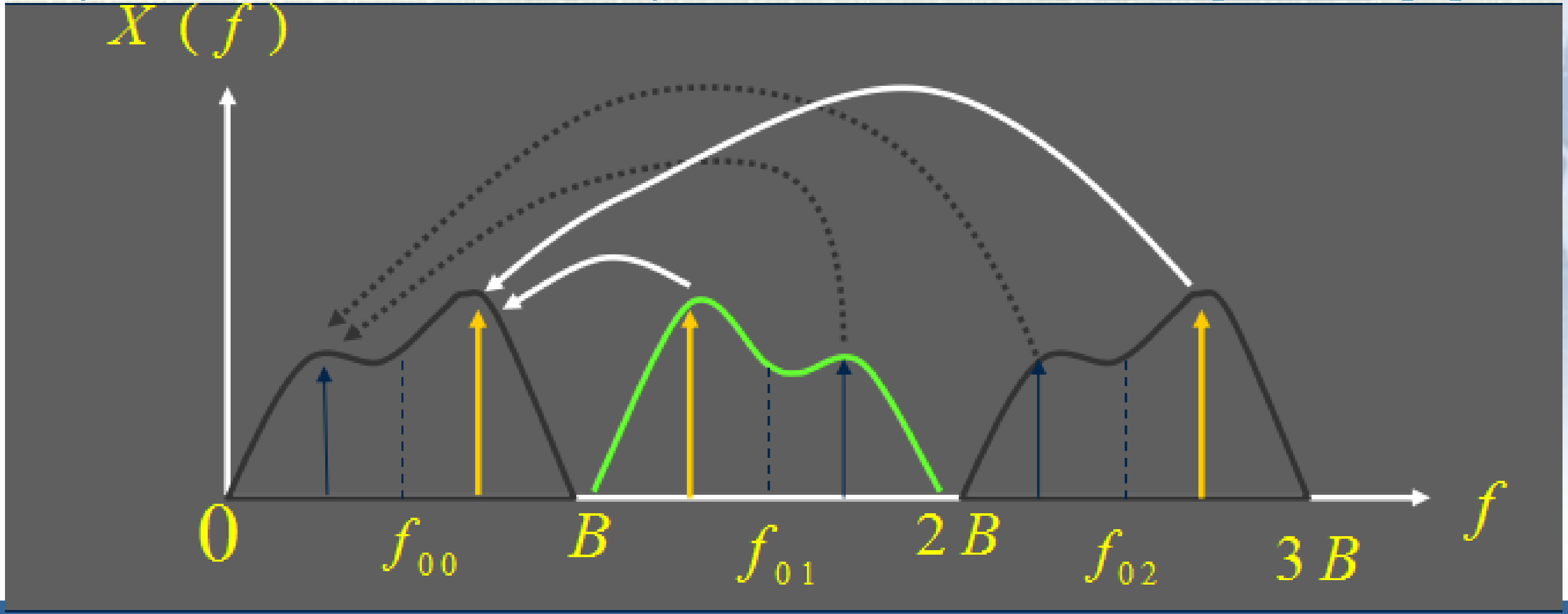
Then, the sampled signal $g_a(nT_s)$ can be used to represent original signal $g_a(t)$.

Note:

- Before band-sampling, the signal in only **one frequency-band** can be permitted.
- For lowest sampling rate, that is $f_s=2B$, the signal center frequency f_0 should be:

$$f_0 = \frac{2n+1}{2}B \quad \text{or} \quad f_L + f_H = (2n+1)B$$

- **The band-sampling result is moving the signal in the frequency band of (nB, (n+1)B)(n=0,1,2,…) to the frequency band of (0, B).**
- **When n is odd number, the relationship between them is reversal; while n is even, the relationship is opposite.**



- As can be seen, $g_a(t)$ can be recovered from $g_p(t)$ by passing it through an ideal bandpass filter with a passband given by $\Omega_L \leq |\Omega| \leq \Omega_H$ and a gain of T.
- Note: Any of the replicas in the lower frequency bands can be retained by passing through bandpass filters with passbands

  $$\Omega_L - k(\Delta\Omega) \leq |\Omega| \leq \Omega_H - k(\Delta\Omega)\ ,\ 1 \leq k \leq M-1$$

  providing a translation to lower frequency ranges.

Reference Homework

- 3.11,3.17, 3.18(a)(c)(e), 3.25, 3.58, 3.62, 3.63, 3.66
- M3.1, M3.2